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1  #include <stdio.h>
2  int main()
3  {
4      int bt[20], p[20], wt[20], tat[20];
5      int i, j, n, total = 0, pos, temp;
6      float avg_wt, avg_tat;
7
8      printf("Enter number of process: ");
9      scanf("%d", &n);
10
11     printf("\nEnter Burst Time:\n");
12     for (i = 0; i < n; i++)
13     {
14         printf("P%d: ", i + 1);
15         scanf("%d", &bt[i]);
16         p[i] = i + 1;
17     }
18     // Sorting (SJF)
19     for (i = 0; i < n; i++)
20     {
21         pos = i;
22         for (j = i + 1; j < n; j++)
23         {
24             if (bt[j] < bt[pos])
25                 pos = j;
26         }
27         // Swap burst time
28         temp = bt[i];
29         bt[i] = bt[pos];
30         bt[pos] = temp;
31
32         // Swap process number
33         temp = p[i];
34         p[i] = p[pos];
35         p[pos] = temp;
36     }
37
38     // Waiting time calculation
39     wt[0] = 0;
40     for (i = 1; i < n; i++)
41     {
42         wt[i] = 0;
43         for (j = 0; j < i; j++)
44         {
45             wt[i] += bt[j];
46         }
47         total += wt[i];
48     }
49
50     avg_wt = (float)total / n;
51     total = 0;
52
53     // Turnaround time calculation
54     printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");
55     for (i = 0; i < n; i++)
56     {
57         tat[i] = bt[i] + wt[i];
58         total += tat[i];
59
60         printf("P%d\t%d\t%d\t%d\n",
61             p[i], bt[i], wt[i], tat[i]);
62     }
63
64     avg_tat = (float)total / n;
65
66     printf("\nAverage Waiting Time = %f", avg_wt);
67     printf("\nAverage Turnaround Time = %f\n", avg_tat);
68
69     return 0;
70 }

```

Enter number of process: 3

Enter Burst Time:

P1: 45

P2: 32

P3: 18

Process	Burst Time	Waiting Time	Turnaround Time
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P3	18	0	18
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P2	32	18	50
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P1	45	50	95
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Average Waiting Time = 22.666666

Average Turnaround Time = 54.333332

=== Code Execution Successful ===